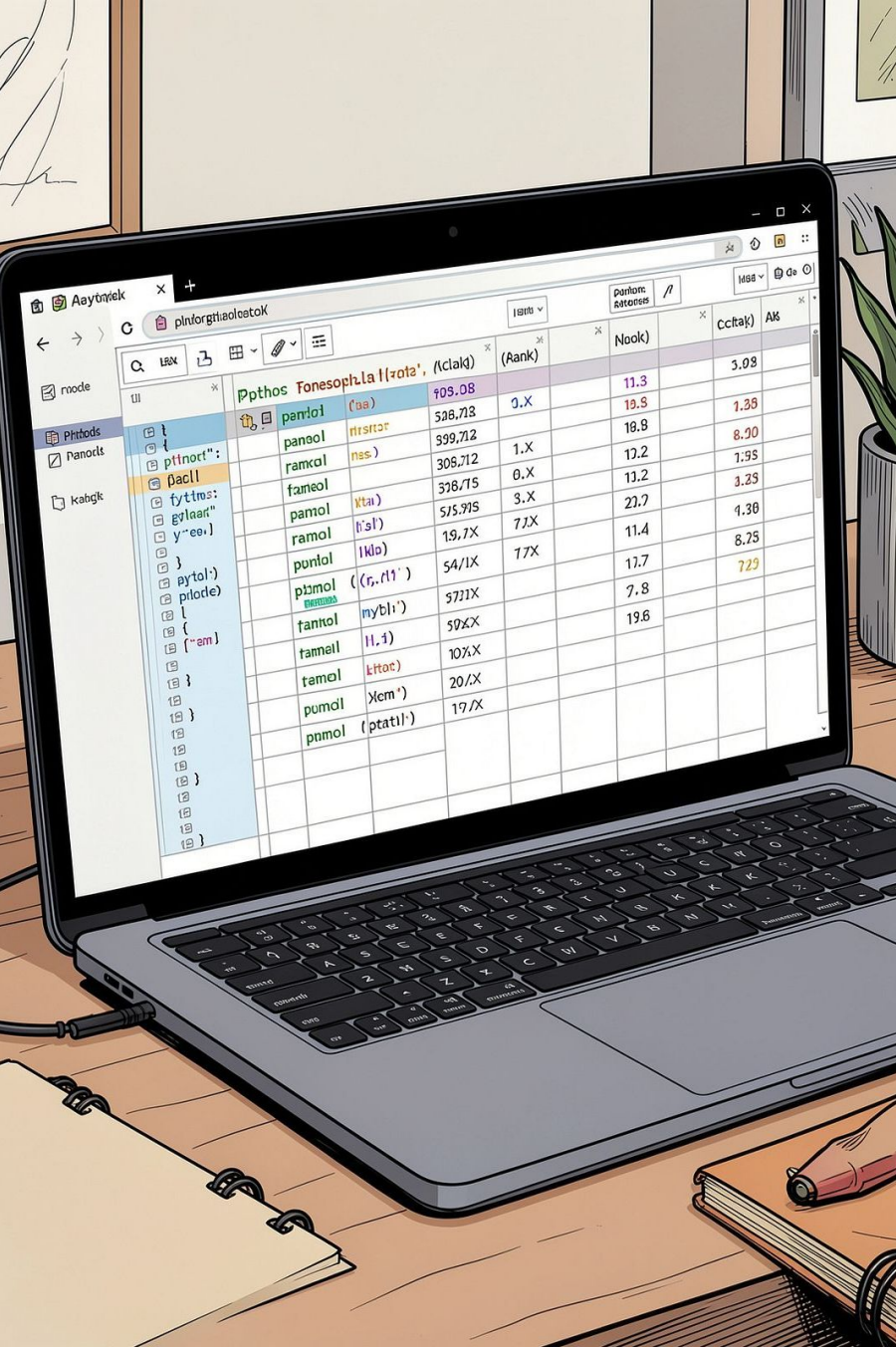


Customer Shopping Behavior Analysis

3,900 purchases • 18 columns • Goal: uncover spending patterns, segments, preferences, and subscriptions





Dataset Summary

Rows & Columns

Rows: 3,900 •
Columns: 18

Key Features

Demographics,
purchase details,
behavior, shipping,
discounts

Missing Data

Review Rating: 37
missing



EDA & Preparation (Python)

01

Load & Inspect

`df.info(), df.describe()`

02

Missing Imputation

Review Rating → median by product category

03

Standardize & Engineer

`snake_case; age_group; purchase_frequency_days`

04

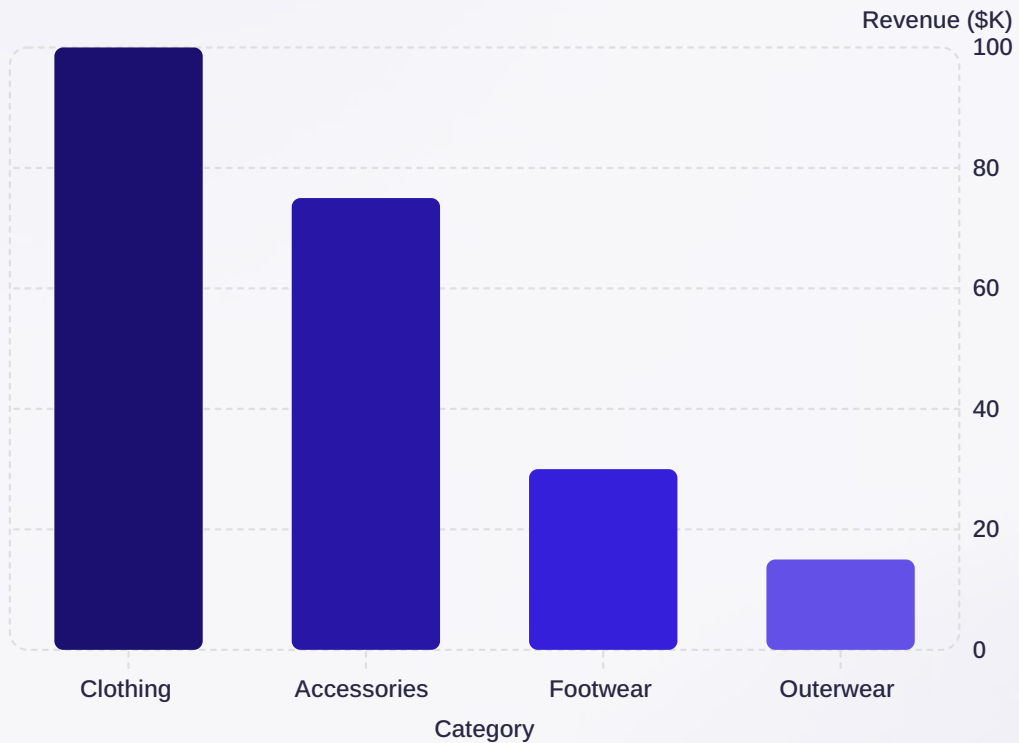
DB Integration

Load cleaned DataFrame to PostgreSQL



SQL Analysis — Key Questions

- Revenue by gender
- High-spending discount users
- Top products by rating
- Shipping type comparison
- Subscribers vs. non-subscribers



Category Performance

Revenue and sales are concentrated in Clothing and Accessories. Use this for campaign focus.

📄 Sales: Clothing 1,800 • Accessories 1,200 • Footwear 600 • Outerwear 200



Shipping & Spend

Standard

Average purchase:
\$58.46

Express

Average purchase:
\$60.48

Express users slightly outspend standard users → consider targeted offers

Customer Segmentation



Loyal

3,116 customers



New

83 customers

Returning

701 customers



Dashboard Highlights

- Customers: 3.9K
- Average purchase: \$59.76
- Average review rating: 3.75
- Subscribers: 27% • Non-subscribers: 73%

☐ Visuals: revenue by category, sales by category, age group breakdowns

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow the loyal segment



Review Discounts

Balance sales uplift with margin control



Product Positioning

Promote top-rated and best-selling products



Targeted Marketing

Focus on high-revenue age groups and express-shipping users